

Tic Tac Toe online multiplayer game

Motivation for why this is an interesting project

- Tic Tac Toe is a classic game, and developing an online version adds an interesting twist. It not only involves creating a graphical user interface (GUI) for the game but also introduces networking to allow two users to play against each other. This project provides an opportunity for us to delve into GUI programming, networking, and multithreading-meeting the objectives outlined in the project description.

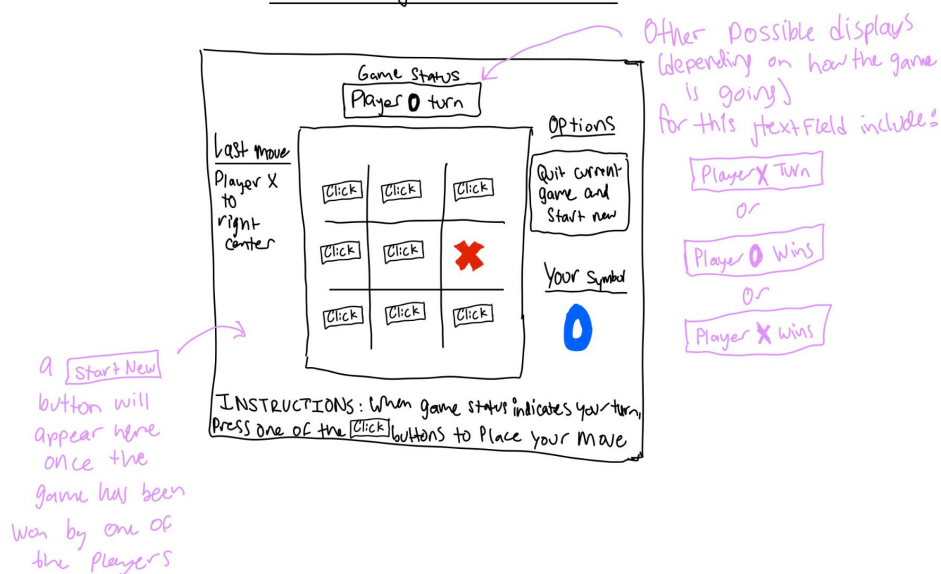
Description of what task/problem you're trying to solve

- Our project aims to create a multiplayer online Tic Tac Toe game where two users can play against each other over a network. This involves designing a user-friendly interface, implementing game logic, and incorporating networking to facilitate real-time gameplay.

Description of the GUI along with with a sketch

- The GUI will have a design with a 3x3 grid representing the Tic Tac Toe board. Each cell in the grid can be clicked to place the X or O symbol (depending on which symbol the user is randomly assigned to at time of game). There will be an area displaying the current game status, indicating whose turn it is or if the game has been won by either player. Additionally, the GUI will include a button to quit the game early and begin a new one, a button to start a new game after the game has been won, a JTextArea indicating the last move played on the gameboard, an area indicating which symbol the user is assigned, and instructions listed at the bottom.

Final Project GUI Sketch



Description of how your project incorporates networking and/or multithreading

- The game will use a client-server model. The server will keep track of the game's state and send updates to the clients when a player makes a move. The clients will then update their GUIs to reflect the new game state.

Description of how and why the project is sufficiently challenging

- The project is challenging as it involves the integration of GUI programming, networking, and multithreading. Managing the real-time communication between players, ensuring a synchronized game state, and handling user interactions add complexity to the code development process. Additionally, creating an engaging and visually appealing GUI requires attention to design principles.

Description of a testing plan and its deliverables

- Testing Plan: The testing plan for the Tic Tac Toe online game will cover various aspects, ensuring the reliability and functionality of the application. The plan will include both manual testing by the team and automated testing where applicable.
 - GUI Responsiveness:
 - Test user interactions such as clicking on cells, starting a new game, and exiting the application.
 - Check for consistent display and updates of the game status.
 - See if GUI accurately displays which players turn it is, what the last move was, and that both players were randomly assigned unique symbols (avoid accidentally randomly assigning both players 'X')
 - Network Connectivity:
 - Test the ability to establish and maintain a connection between two players.
 - Game Logic Accuracy:
 - Verify that the game follows the rules of Tic Tac Toe.
 - Test different move scenarios to ensure accurate gameplay:
 - Confirm that the game correctly identifies a win, loss, or draw.
 - Concurrency and Multithreading:
 - Test simultaneous moves from both players to ensure synchronization.

A description of how the manager's workload will be adjusted to account for the extra work they are doing

- Alina, serving as the group manager for this project, holds responsibilities such as establishing the initial GitHub repository, coordinating group meetings, and ensuring the

punctual submission of project deliverables. To ensure an equitable distribution of tasks among team members, we will engage in weekly discussions to determine how to adjust workloads, specifically addressing managerial responsibilities. This ongoing dialogue will allow us to identify and modify aspects of Alina's workload over the course of several weeks, ensuring fairness and efficiency throughout the project's duration.